Knowledge Graph Quality Evaluation

AnimalRAG GraphRAG knowledge graphs, all 38 topic groups + test set

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1. Executive summary

This report evaluates the triplet-level quality of the knowledge graphs produced by the AnimalRAG GraphRAG pipeline, following Prof. Tao's directive to verify triplet accuracy before building further on the graphs. Two evaluations were run: a deep manual verification of a random triple sample against the full text of the source papers (group13), and an automated structural and consistency analysis of every exported graph (83,554 triples across 38 group graphs and the test-set graph).

Finding	Value
Triples analyzed (all graphs)	83,554
Judged accuracy, group13 sample of 50 (fully correct)	58%
imprecise (right idea, wrong granularity / dropped condition)	26%
wrong (contradicted, inverted, garbled, context leak)	16%
Distinct relation types across all graphs	7,603 (4,267 used exactly once)
Trivial relation triples (is measured in, is component of, ...)	12,705 (15.2%)
Unqualified contradiction pairs (same subject & object, opposing relations)	495 - present in every group (2 to 46 per group)
Duplicate s-r-o triples / self-loops	0 / 0 (entity merging works)
Triples supported by 2+ papers	315 (0.4%)

Key message: the pipeline runs end to end and its entity merging is clean, but roughly one triple in six in the judged sample is wrong, the relation vocabulary has exploded, and every graph contains pairs of directly opposing, unqualified claims. None of these problems is visible to answer-level evaluation (the RESULTS.md faithfulness criterion was a tie 83% of the time), which is exactly why triple-level verification has to come first. All of them are fixable with a judge filtering stage, relation normalization, and pruning - without changing the extraction pipeline.

1b. Overall assessment: is the KG good?

Short answer: it is a good DRAFT knowledge graph - strong engineering, not yet a trustworthy fact store. Both halves matter.

What is genuinely good

- Scale and robustness: 39 graphs built end to end, resumable, cached, with full manifests - this infrastructure is the hard part and it works.
- Entity merging is clean: zero duplicate triples and zero self-loops across all 83,554 triples. Many published pipelines fail exactly here.
- Full provenance: every triple stores its source papers and chunk ids. This single design decision is what makes every fix below mechanical rather than a rebuild.
- Rich schema: effect direction, weights, contexts and descriptions per triple - ahead of what most text-to-KG papers export.

What is not yet good enough for GraphRAG

- Accuracy: ~1 in 6 sampled triples is wrong, and these errors are invisible at answer level (faithfulness judging tied 83% of the time). A KG-grounded answer can currently cite a false fact with full confidence.
- Relation noise: 7,603 relation types, 56% used once - retrieval over relations becomes string matching instead of graph semantics.
- Contradictions: 495 unqualified opposing pairs means the same question can be answered both ways depending on which triple retrieval surfaces.

Important context: every failure class found here is the documented failure mode of naive LLM extraction (noise, condition dropping, hallucinated or inverted relations - exactly the three problems the GraphJudge paper, EMNLP 2025, was built to solve). This is the expected state of a first-pass extraction, not a design flaw. The pipeline was built verification-ready; the verification layer is simply the next component.

2. What was evaluated and how

2.1 Data

- All readable graph exports from AnimalRAG_Local-main/knowledge_graph/stores/<group>/export/triples.csv: 38 group graphs + the test-set graph, 83,554 triples in total. Row counts match RESULTS.md exactly, confirming the mapping.
- The combined all-papers graph (setting B) exists as a 35 MB triples.csv (modified Oct 8) but exceeds the 10 MB Drive-connector download limit; the identical analysis script is included so it can be run locally on that file in minutes.

2.2 Manual triple verification (deep dive)

- Group13 (Iodine_Thyroid_Content, 3 papers, 474 triples): a seeded random sample of 50 triples was judged against the FULL TEXT of all three source papers (Schone 2006; Li et al. 2012; Rottger et al. 2011), each read end to end.
- Verdicts: correct / imprecise / wrong, each with a written evidence comment quoting the source or naming the error. The per-triple verdict file is available for audit (sample50_judged.csv).
- The GraphJudge evaluation code (EMNLP 2025) was run on the sample against the judged-correct reference, giving a triple precision of 0.58.

2.3 Automated analysis (all graphs)

- Structure: triple counts, relation vocabulary size, singleton relations, duplicates, self-loops, vague entities, trivial-relation share, cross-paper support.
- Consistency: contradiction census - pairs with the same subject and object where one relation is an explicit negation of the other (affects vs has no effect on) or a direction conflict (increases vs decreases/reduces), with no qualifying context on either triple.

3. Deep dive: judged accuracy (group13)

Verdict	Count / 50	Share
Correct (supported by source text)	29	58%
Imprecise (granularity or condition lost)	13	26%
Wrong (contradicted / inverted / garbled)	8	16%

Error taxonomy with examples from the graph

- Object truncation: 'high dietary iodine increases blood serum' - the source says it increases the serum IODINE CONCENTRATION. Also 'KI increases liver', 'iodine supplementation increases iodine'.
- Condition dropping: 'iodine causes hypothyroidism' (source: EXCESS iodine); 'iodine poses no risk to human health' (source: no risk FROM PORK at these feed levels).
- Surface inversion: 'rape reduces thyroid gland' - the paper shows rapeseed ENLARGES the thyroid while reducing its iodine stores.
- In-graph contradiction: 'iodine affects fattening performance' and 'iodine does not affect fattening performance', both extracted from the same paper.
- Citation-context leak: 'swine feed improves human iodine supply' is a cited aim in the introduction; the paper's own conclusion is that pork's contribution is marginal.

- Garbled: 'potassium iodide is source of milk'; 'spleen is part of empty body' (the source explicitly defines empty body as the carcase without innards).

4. All groups: structural and consistency results

Columns: triples; distinct relations; share of relations used only once; share of triples with trivial relations; contradiction pairs found. Duplicates and self-loops are zero everywhere and omitted.

Group	Triples	Relations	Singleton rel.	Trivial triples	Contradictions
g01 Review_Dairy_Based	2868	670	56%	14.1%	10
g02 Chicken_Calcium_Enriched	2864	604	59%	19.0%	20
g03 Model_Production_Male	2510	505	62%	12.0%	24
g04 High_Dairy_Host	2416	524	58%	23.6%	9
g05 Dairy_Impact_Goats	3535	733	56%	9.4%	7
g06 Selenium_Nutrition_Livestock	2738	667	61%	15.0%	9
g07 Biohydrogenation_Unsat_Fatty	2858	657	55%	10.2%	23
g08 What_Cryptoxanthin_Vitamin	982	317	57%	10.8%	5
g10 Cattle_Influence_Meat	2094	479	58%	11.6%	11
g11 Alfalfa_Intake_Cattle	995	199	60%	12.0%	21
g12 Feed_Poultry_Additives	2660	437	54%	7.7%	29
g13 Iodine_Thyroid_Content	474	127	50%	30.8%	13
g14 Acid_Milk_Fatty	2491	431	56%	21.1%	28
g15 Nutrition_Human_Novel	2842	647	60%	13.8%	7
g16 Nutrition_Human_Products	2513	548	58%	18.4%	5
g17 Health_Human_Exposure	3262	654	54%	18.9%	8
g18 Health_Human_Exposure2	2257	509	57%	9.7%	9
g20 Feed_Livestock_Feeds	2118	515	60%	12.3%	6
g21 Intestinal_Microbiota_Pigs	3250	591	56%	7.3%	12
g22 Gut_Supplementation_Soluble	465	152	56%	10.5%	11
g23 Acid_Fatty_Oil	2163	472	52%	20.7%	27
g24 Acid_Cla_Egg	1273	311	62%	9.3%	11
g25 Amino_Acid_Ileal	1362	279	58%	27.8%	11
g26 Food_Human_Chain	3132	645	55%	21.4%	5
g27 Meat_Quality_Factors	2069	405	53%	19.9%	3
g28 Feed_Human_Moringa	2398	541	58%	13.4%	14
g29 Performance_Growth_Digest	2686	395	55%	13.6%	46
g30 Performance_Digestive_Enzymes	587	108	49%	12.3%	16
g31 Dairy_Farms_Food	960	225	53%	27.1%	2
g32 Implications_Human_Health	1569	382	59%	15.0%	2
g33 Pig_Antioxidant_Meta	2557	606	56%	16.3%	10
g34 Pig_Bovine_Metagenomic	1791	457	60%	10.2%	6
g35 Properties_Nutritional_Pot	3073	575	56%	16.5%	5
g36 Farm_Animals_Lactation	1766	398	58%	19.5%	4
g37 Carcass_Meat_Performance	1913	272	56%	14.5%	26
g38 Nutritional_Value_Meat	1321	266	53%	16.7%	5
g40 Potential_Application_Garlic	1612	338	57%	14.4%	16
g41 Organic_Acids_Essential	1633	334	56%	4.8%	14

Group	Triples	Relations	Singleton rel.	Trivial triples	Contradictions
testset	3497	672	61%	20.9%	5

Totals: 83,554 triples; 7,603 distinct relations (56% singletons); 12,705 trivial triples (15.2%); 495 contradiction pairs. Most-used relations: contains (5,514), increases (4,928), improves (3,971), reduces (3,597), is source of (3,582).

5. Contradiction census

A contradiction pair is two triples with the same subject and object whose relations directly oppose each other, with no qualifying context (dose, species, duration) stored on either triple. Some reflect genuinely conflicting findings across papers - but stored this flat, a GraphRAG query can retrieve either side as 'the' fact. 495 such pairs exist; examples:

Group	Subject	Relation A	Relation B	Object
g37	sodium monensin	increases	reduces	feed efficiency
g41	essential oils	increases	decreases	feed intake
g04	probiotic	increases	reduces	average daily feed intake
g29	soybean lecithin	has no effect	increases	body weight
g14	forage lucerne	has no effect on	affects	milk yield
g03	white lupin	does not influence	influences	hind leg meat
g28	moringa oleifera	increases	reduces	alpha-tocopherol
g23	n-6:n-3 ratio	increases	decreases	thigh muscle
g37	sunflower oil	does not affect	influences	growth performance
g07	ensiled olive cake	has no effect on	increases	fatty acid synthase

Note the first row: monensin and feed efficiency is directly adjacent to the team's own QA test question about monensin. A user asking 'does monensin improve feed efficiency?' can be served either triple.

6. How to improve it: phased plan

Ordered by effort-to-impact. Phases 1-2 need no LLM and no changes to extraction; they can run this week.

Phase	Action	Effort	Expected impact
1	Prune trivial triples (is measured in, is component of, ...) and normalize the 4,267 singleton relations onto a controlled vocabulary (~50 relations)	Script only, 1-2 days	Graph shrinks ~15%, relation space shrinks >50%, retrieval precision up with zero information loss on facts
2	Contradiction triage: for each of the 495 pairs, pull both triples' source chunks (chunk_ids already stored) side by side for review	Script + review sheet, 2-3 days	Separates extraction errors (delete) from genuine conflicting findings (qualify) - also produces labeled data for Phase 3
3	Judge filter stage: run the released GraphJudge LoRA judge (no training needed) over every triple vs its own source chunk; then fine-tune a domain judge on HPRC using Phase 2 labels + gold set	1-2 weeks, HPRC in hand	Targets the 16% wrong class; paper shows >90% judgement accuracy; precision rises without recall loss (unlike rule filters)
4	Qualified triples: attach condition fields (dose, species, duration, study) to effect relations so conflicting findings coexist correctly	Schema + re-extraction prompt change	Fixes the root cause of contradictions; this is also the team's publishable research contribution
5	Re-evaluate: re-run this analysis plus the RAG-vs-GraphRAG benchmark on the cleaned graph (and setting B)	Automated, ~2 days GPU	Measures whether KG quality gains show up as answer quality gains - the result both papers need

Summary recommendations

- **1. Add a graph-judgement filter stage (GraphJudge, EMNLP 2025): every triple is checked against its own source chunks, which the export already stores (chunk_ids column). This targets the 16% wrong class mechanically; released LoRA judge weights exist, and HPRC allocation is in hand for a domain fine-tune.**
- 2. Normalize relations: 7,603 relation types is OpenIE-style vocabulary explosion. Mapping singletons onto a controlled vocabulary (or clustering, as in KGGen) would shrink the relation space by more than half without losing facts.
- 3. Prune trivial triples: 12,705 'is measured in / is component of' style triples add retrieval noise with near-zero knowledge value. This is the concrete, no-LLM-needed starting point for the pruning work.
- 4. Resolve or qualify contradictions: for each of the 495 pairs, either one side is an extraction error (judge decides) or both findings are real and need their conditions (dose, species, study) attached. The second case is the 'qualified triples / richer knowledge units' research direction - our corpus now provides the evidence that flat triples cannot represent it.
- 5. Re-run this analysis on the 35 MB all-papers graph (setting B) locally - the script is in the repo and runs in minutes; cross-paper merging will likely raise the contradiction count and makes qualification more urgent.

7. Files and reproducibility

- Repo: github.com/ssrikar1503/directed-studies-kgcompass -> experiments/sumanth-kg-eval/
- all-groups/per_group_quality.csv - the full per-group table; all-groups/contradictions.csv - all 495 pairs with group, subject, relations, object.
- sample50_judged.csv - the 50 judged triples with verdicts and quoted evidence, open for audit.
- SUMANTH_KG_EVAL.md - the group13 deep-dive write-up; the analysis script reproduces every number in this document from the triples.csv exports.